

2.4 程序调试器gdb¹

2.4.1 gdb简介²

gdb是GNU Project Debugger的简称，也是GNU(GNU's Not Unix)开源项目中遵循GPL发行许可证的free software。掌握gdb调试技巧，在项目上遇到问题才会得心应手。gdb 官方网站 (<https://sourceware.org/gdb/>)³

一般来说，GDB 主要能够提供以下四个方面的帮助：⁴

GDB can do four main kinds of things (plus other things in support of these) to help you catch bugs in the act:⁵

- Start your program, specifying anything that might affect its behavior. (指定一些参数)⁶
- Make your program stop on specified conditions. (断点)⁷
- Examine what has happened, when your program has stopped. (分析crash现场)⁸
- Change things in your program, so you can experiment with correcting the effects of one bug and go on to learn about another. (直接修改程序，看结果)⁹

GDB: The GNU Project Debugger¹⁰

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GDB: The GNU Project Debugger



What is GDB?¹¹

GDB, the GNU Project debugger, allows you to see what is going on 'inside' another program while it executes -- or what another program was doing at the moment it crashed.¹²

GDB can do four main kinds of things (plus other things in support of these) to help you catch bugs in the act:¹³

- Start your program, specifying anything that might affect its behavior.¹⁴
- Make your program stop on specified conditions.¹⁵
- Examine what has happened, when your program has stopped.¹⁶
- Change things in your program, so you can experiment with correcting the effects of one bug and go on to learn about another.¹⁷

Those programs might be executing on the same machine as GDB (native), on another machine (remote), or on a simulator. GDB can run on most popular UNIX and Microsoft Windows variants, as well as on macOS.¹⁸